Rishi Vora

BS-MS in Physics from IISER Mohali

vorarishi22@gmail.com | Portfolio | GitHub | LinkedIn

Education

Indian Institute of Science Education and Research (IISER) 2021 — 2026
Integrated Bachelors & Masters, with Major in Physics and Minor in Data Science *Mohali*

- **CPI:** 8.0 / 10
- **Relevant Courses:** Foundations of Quantum Mechanics and Quantum Information | Advanced Quantum Mechanics | Introduction to Data Science | Machine Learning and Artificial Intelligence | Computational Methods in Physics | Algorithms and Complexity | Nonlinear Dynamics and Chaos | Modelling Complex Systems

Research Experience

Quantum generative machine learning for state generation 12/2025 — 09/2026
Master's Thesis *IISER Pune*

- **Guide:** Prof. Santhanam | **Code:** github.com/Vortriz/thesis
- Created a new quantum state generation model that requires fewer resources and has significantly faster convergence compared to existing models, such as QuDDPM.
- Applied it to various tasks, such as learning many-body phases, states distributed over nontrivial topology, localized states of Quantum Kicked Rotor etc.
- Benchmarked gradient-free optimization strategies such as QNSPSA, SPSA, and Rotosolve against gradient-based approaches.
- Created a Julia implementation using Yao.jl, with optional CUDA acceleration.

Reading project on QKD protocols Summer 2024
Summer reading project *IISER Mohali*

- **Guide:** Prof. Arvind
- Reviewed working of BB84, B92 and E91 QKD protocols.
- Worked out Shor and Preskill's security proof of BB84 protocol.

Quantum algorithms and circuit decomposition Summer 2023
Summer internship *IISER Mohali*

- **Guide:** Dr. Sandeep Goyal | **Code:** github.com/Vortriz/Summer2023
- Implemented Full Adder, Multiplier, Superposition Adder, and XOR adder circuits in Qiskit.
- Attempted a circuit decomposition scheme using Solovay-Kitaev algorithm.

Skills & Certifications

Qiskit Global Summer School 2024 2024
IBM

- Certificate of Quantum Excellence
- Lectures, interactive sessions, and programming tasks on Circuit Transpilation, Modelling and Characterization of Hardware Noise, Execution on Noisy Quantum Hardware using Qiskit.

Qiskit Global Summer School 2023 2023
IBM

- Certificate of Quantum Excellence
- Lectures, interactive sessions, and programming tasks on Dynamic circuits, Phase estimation, Factoring, Error mitigation, VQE, and Iterative phase estimation.

Programming & Tools – **Python** (Qiskit, PyTorch, TensorCircuit) | **Julia** (Yao.jl) | **Linux** (NixOS) | **Git** | **Typst** | Code written as a part of the coursework can be found on my website.

Spoken Languages – **English** (Fluent) | **Hindi** (Native) | **Gujarati** (Native)

Hobbies – I actively develop and contribute to FOSS projects on GitHub, create websites once in a while, and spend my downtime listening to music and watching eclectic films.

Scholarships

IISER Mohali — Merit-Cum-Means Scholarship 2022 — 2026

Bi-annual scholarship supporting Bachelor's and Master's studies.

Extracurriculars

Won 2 bounties in unitaryHACK 2026 June 2026
Unitary Foundation

- Annual virtual hackathon organized by Unitary Foundation for contributing to open-source quantum ecosystem.
- Bounties:
  - `quantumsavory.jl` – `show` and visualization methods for multipartite Gaussian states from Gabs.
  - `paulipropagation.jl` – Embed example Jupyter notebooks into docs website.

Conducted Workshops on Python Tooling and Typst March 2026
IISER Pune

- **Link**: github.com/Vortriz/workshops-and-presentations
- Conducted interactive workshops focused on addressing problems with Python tooling ecosystem and showcasing Typst for high-quality document preparation.

Convenor of Turing Club (Computation Club) 2023 — 2024
IISER Mohali

- Organized Hackathons, Workshops, Talks etc.
- Took programming help sessions.